

# Ziya Sherzeen S

## Artificial Intelligence and Data Science Engineer

### Education

| Year      | Degree                                             | Institute Name                   | Marks |
|-----------|----------------------------------------------------|----------------------------------|-------|
| 2022-2026 | B. Tech (Artificial Intelligence and Data Science) | Karpagam Institute of Technology | 8.29  |

### Contact Info

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**Leetcode:** [https://leetcode.com/u/Ziya\\_sherzeen/](https://leetcode.com/u/Ziya_sherzeen/)

**LinkedIn:** <https://www.linkedin.com/in/ziya-sherzeen/>

**GitHub:** <https://github.com/ZiyaSherzeen>

### Career Objective

An Artificial Intelligence and Data Science engineering student with a strong foundation in Machine Learning and full-stack development. Passionate about crafting intelligent, data-driven solutions that integrate analytical depth with practical web technologies. Eager to contribute to innovative projects that solve real-world problems through the synergy of AI and modern software engineering.

### Projects

- 1.Vehicle Routing Problem | Guided Project** (Jan 8-Jan 9)  
**Objective:** Optimized the Vehicle Routing Problem using Genetic Algorithms in DEAP, minimizing travel distance and time.
  - Utilized Genetic Algorithms in DEAP with a distance matrix and time-based fitness evaluation.
  - Achieved minimized travel time and distance, ensuring efficient and effective routing.
- 2.Dimensionality-Reduction-Using-K-Means-for-Activity-Recognition | Guided Project** (Jan 10)  
**Objective:** Implemented K-Means clustering for dimensionality reduction in activity recognition, improving feature selection and model efficiency.
  - Used K-Means clustering to identify and group redundant features in the activity recognition dataset.
  - Trained a model using the reduced feature set, improving computational efficiency and maintaining accuracy.
- 3.Market Segmentation Using DBSCAN | Guided Project** (Mar 9)  
**Objective:** Segment customers based on spending behaviour and purchase frequency using DBSCAN, helping businesses with targeted marketing and customer retention.
  - DBSCAN dynamically determines clusters instead of requiring a fixed number.
  - Effectively segments **diverse customer groups** without assuming uniform distribution.
  - Helps businesses **personalize offers, optimize marketing, and improve customer engagement**.
- 4.HealthTech – AI driven diabetes analysis | Guided Project** (Mar 11)  
**Objective:** Developed a machine learning model to analyse diabetes patient readmission patterns using the UCI Diabetes dataset, focusing on key health indicators, medication changes, and predictive analytics.
  - Handled missing values with **KNN Imputation** and standardized features.
  - Trained **Logistic Regression & Random Forest** for readmission prediction.
  - Analysed **HbA1c impact, medication changes, and readmission trends** with **heatmaps & ROC curves**.
- 5. Typing Speed Test Web Application - A real-time typing performance evaluation | Self-Project** (May 28)  
**Objective:** To develop a web-based application that accurately measures typing speed and accuracy.
  - Designed and implemented a responsive interface using **HTML, CSS, and JavaScript**.
  - Features include dynamic **paragraph generation**, real-time **WPM calculation**, and **error tracking**.
- 6. Kanban Board - Task Management Web App | Self-Project** (June 25-June 27)  
**Objective:** Developed a Kanban board using React for task management using **HTML, CSS, JavaScript, React**.
  - Added drag-and-drop, edit, and delete features.
  - Implemented visual drop indicators and a trash zone for intuitive task removal.

### Certificates

- Certified AWS Academy Cloud Foundations Graduate
- Certified in RDBMS PostgreSQL through the Spoken Tutorial Project by IIT Bombay, upon successful completion of the training and online examination
- Completed the MongoDB certification for Document Modelling on the official MongoDB platform.

### Skills

**Programming Languages:** Python, Java, C

**Web Development:** HTML, CSS, JavaScript, React Js

**Machine Learning:** KNN, Genetic Algorithm, DEAP, NLP, DBSCAN

**Soft Skills:** Critical Thinking, Team Collaboration, Communication Skills, Leadership

### Achievement

**Selected for Ideathon Round – Girl Hackathon 2025 by Google**

Qualified for Round 2 of Google's nationwide Girl Hackathon (SWE Track), based on performance in the initial selection round (Problem-Solving).